Experiment-7

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1. Consider a relation R having attributes as R(ABCD), functional dependencies are given below:
AB->C, C->D, D->A
Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Ans:

Closure:

B+= B
BA+= BACD
BC+= BCDA
BD+= BDAC

- Prime Attributes = {A,B,C,D}
- Super set / Candidate set = {AB, BC, BD}

Since, all the RHS attributes are prime, the relation R is in 3NF.

Check for BCNF:
For BCNF, for every non-trivial FD X -> Y, X must be a superkey.
C -> D, C is not a superkey, therefore it violates BCNF.

Highest Normal Form: 3NF

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2. Relation R(ABCDE) having functional dependencies as :
A->D, B->A, BC->D, AC->BE
Identify the set of candidate keys possible in relation R. Find the highest normal form.

Ans:

Closure:

C+=C
CA+=CADBE
CB+=CBAED
CD+=CD
CE+=CE

- Candidate sets = {CA, CB}
- Prime attributes = {A, B, C}

Since in A->D, D can be determined by only A (subset of candidate key AC), it violates partial dependency so it is not in 2NF.

Highest Normal Form: 1NF

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3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:
B->A, A->C, BC->D, AC->BE
Identify the set of candidate keys possible in relation R. Find the highest normal form.

Ans:

Closure:

B+=BACDE
A+=ACBED
C+=C
E+=E

D+=D

- Candidate keys: {A, B}
- Primary keys: {A, B}

Since all the attributes on left side are either candidate keys or super keys, hence the relation R is in BCNF form.

Highest Normal Form: BCNF

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4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:
A->BCD, BC->DE, B->D, D->A
Find the highest normal form.

Ans:

F never appears on the RHS of any FD, therefore it must be present in every candidate key.

AF+=AFBCDE
BF+=BFDACE
CF+=CF
DF+=DFABCE
EF+=EF

- Candidate keys: {AF, BF, DF}
- Prime attributes: {A, F, B, D}

A -> **C**
A is a proper subset of candidate key AF, and C is non-prime.
Therefore, there is a partial dependency, so the relation violates 2NF.

Highest Normal Form: 1NF

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5. Relation R(ABCDE) having dependencies as:
CE -> D,
D ->B,
C -> A
Highest normal form? C.K set?

Ans:

Closure:

CE+=CEDBA

- Candidate key = {CE}
- Prime keys = {C, E}

From FD, C -> A
C is a proper subset of candidate key CE, and A is non-prime.
Therefore, C → A is a partial dependency which violates 2NF.

Highest Normal Form: 1NF

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6. Relation R(ABCDEF) having dependencies as:

AB -> C
DC -> AE
E -> F

Highest normal form? C.K set?

Ans:

Since B, D never occur on RHS of any FD, they will be a part of candidate key.
Closure:

BD+=BD
BDA+=BDACEF
BDC+=BDCAEF
BDE+=BDEF
BDF+=BDF

- Candidate keys= {BDA, BDC}
- Prime attributes= {B, D, A, C}

Check for 2NF:

- DC -> AE: A is prime but E is non-prime

Here, DC is a proper subset of candidate key BDC.
Therefore DC -> E is a partial dependency so 2NF is violated.

Highest Normal Form: 1NF

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7. Relation R(ABCDEFGHI) having dependencies as:

AB -> C
BD -> EF
AD -> GH
A -> I

Highest normal form? C.K set?

Ans:

Since A, B, D never occur on RHS of any FD, they will be a part of candidate keys.
Closure:

ABD+=ABDCEFGHI

- Candidate set = {ABD}
- Primary attributes = {A, B, D}

Check for 2NF:

- AB -> C: AB is a proper subset of ABD and C is non-prime. Therefore:

AB -> C is a partial dependency, so the relation violates 2NF.

Highest Normal Form: 1NF

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8. Relation R(ABCDE) having dependencies as:

AB -> CD
D -> A
BC -> DE

Highest normal form? C.K set?

Ans:

Since B doesn't appear on RHS, it must be the part of every candidate key.
Closure:

BC+=BCDEA
AB+=ABCDE
BD+=BDACE
BE+=BE

- Candidate key={BC, BA, BD}
- Prime attributes= {B, C, A, D}

Check for 3NF:
- **AB -> CD:** AB is a super key
- **D -> A:** D is not a super key but A is a prime attribute
- **BC -> DE:** BC is a super key

Check for BCNF:
- **D -> A:** D is not a super key

Therefore it violates BCNF

Highest Normal Form: 3NF

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9. Relation R(ABCDE) having dependencies as:
BC -> ADE
D -> B
Highest normal form? C.K set?

Ans:

Since C does not appear on RHS, it must be a part of candidate key
Closure:

C+=C
BC+=ADEBC
CD+=CDBAE
CE+=CE

- Candidate keys={BC, CD}
- Prime attributes={B,C,D}

Check for 3NF:
- **D -> B:** D is a not a super key, but B is a prime attribute
- **BC -> ADE :** BC is a super key

Check for BCNF:
- **D -> B:** D is a not a super key, it violates BCNF

Highest Normal Form: 3NF

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10. Relation R(ABC) having dependencies as:

A -> B

B -> C

C -> A

Highest normal form? C.K set?

Ans:

Closure;

A+=ABC
B+=ABC
C+=ABC

- Candidate keys={A,B,C}
- Prime attributes={A,B,C}

Since all the LHS attributes are super keys,
Highest Normal Form= BCNF

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11. Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

F1 -> F3

F2 -> F4

(F1,F2) -> F5

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Ans:

Since F1, F2 doesn't appear on RHS, they must be a part of candidate key.
Closure:

F1F2+=F1,F2,F3,F4,F5

- Candidate set={F1F2}
- Prime attributes= {F1, F2}

Check for 2NF:

- F1 -> F3: F1 is a subset of candidate key F1F2 and F3 is a non-prime attribute, therefore it violates 2NF
- F2 -> F4: F2 is a subset of candidate key F1F2 and F4 is a non-prime attribute, therefore it violates 2NF

Highest Normal Form: 1NF